Figure (c) shows the calculation for the third section of the output signal, where t > 1. Here the overlap occurs between $\tau = 0$ and $\tau = 1$, making the calculation the same as for the second segment, except a change to the limits of integration:

$$y(t) = \int_0^1 1 \cdot \alpha e^{-\alpha(t-\tau)}\, d\tau \qquad \text{(plug into Eq. 13-1)}$$

$$y(t) = e^{-\alpha t}\, [e^{\alpha\tau}]\Big|_0^1 \qquad \text{(evaluate the integral)}$$

$$y(t) = [e^{\alpha} - 1]\, e^{-\alpha t} \qquad \text{for } t > 1$$

The waveform in each of these three segments should agree with your knowledge of electronics: (1) The output signal must be zero until the input signal becomes nonzero. That is, the first segment is given by $y(t) = 0$ for $t < 0$. (2) When the step occurs, the RC circuit exponentially increases to match the input, according to the equation: $y(t) = 1 - e^{-\alpha t}$. (3) When the input is returned to zero, the output exponentially decays toward zero, given by the equation: $y(t) = ke^{-\alpha t}$ (where $k = e^{\alpha} - 1$, the voltage on the capacitor just before the discharge was started).

More intricate waveforms can be handled in the same way, although the mathematical complexity can rapidly become unmanageable. When faced with a nasty continuous convolution problem, you need to spend significant time evaluating *strategies* for solving the problem. If you start blindly evaluating integrals you are likely to end up with a mathematical mess. A common strategy is to break one of the signals into simpler additive components that can be *individually* convolved. Using the principles of linearity, the resulting waveforms can be added to find the answer to the original problem.

Figure 13-7 shows another strategy: modify one of the signals in some linear way, perform the convolution, and then undo the original modification. In this example the modification is the *derivative*, and it is undone by taking the *integral*. The derivative of a unit amplitude square pulse is two *impulses*, the first with an area of one, and the second with an area of negative one. To understand this, think about the opposite process of taking the integral of the two impulses. As you integrate past the first impulse, the integral rapidly increases from zero to one, i.e., a step function. After passing the negative impulse, the integral of the signal rapidly returns from one back to zero, completing the square pulse.

Taking the derivative simplifies this problem because convolution is easy when one of the signals is composed of impulses. Each of the two impulses in $x'(t)$ contributes a scaled and shifted version of the impulse response to